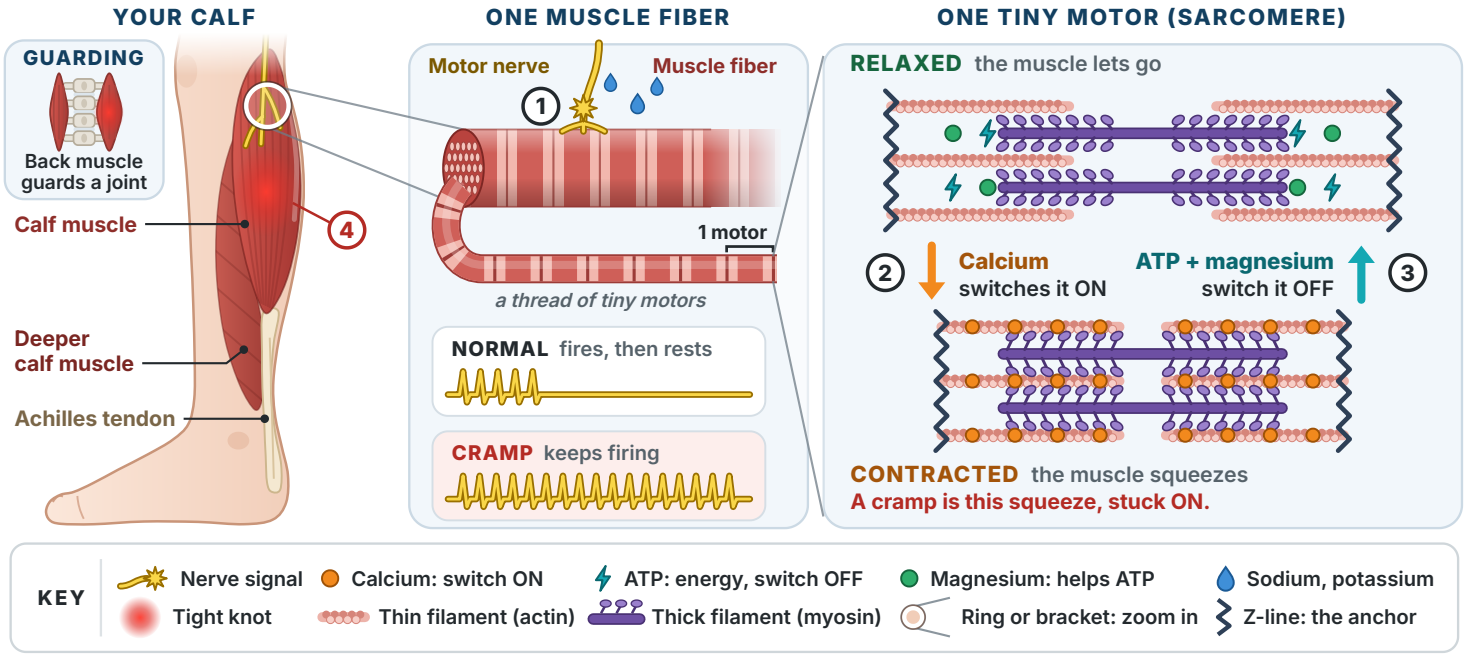
Patient name _____ Date _____

Muscle Cramps and Spasms

Why a muscle locks up, and how to help it let go

A **cramp**, or spasm, is a muscle that suddenly tightens on its own and stays tight for seconds to minutes. It usually grabs the **calf**, foot or thigh, often at night, and it can strike the back or neck too. A spasm can also be a muscle **guarding** a sore back, neck or joint: it tightens to protect the sore spot and is slow to let go. Cramps are very common: **up to 6 in 10 adults** have had a night leg cramp, and nearly **half of people over 60** get them. Most are harmless, and simple habits help.



KEY: Nerve signal · Calcium: switch ON · ATP: energy, switch OFF · Magnesium: helps ATP · Sodium, potassium · Tight knot · Thin filament (actin) · Thick filament (myosin) · Ring or bracket: zoom in · Z-line: the anchor

1. **The nerve says "squeeze."** Sodium and potassium carry the signal into the muscle.
2. **Calcium switches it ON.** Filaments grab and slide. In your blood, enough calcium and magnesium keep nerves calm; too little can set off cramps.
3. **ATP switches it OFF.** With magnesium's help, the filaments let go and calcium goes back into storage.
4. **A cramp is the switch stuck ON.** A tired or over-excited nerve keeps firing.

WHAT MAKES CRAMPS MORE LIKELY

- Tired, overused muscles
- Sweat, heat, too little fluid
- Sitting still for hours
- Injury or a sore joint
- Getting older
- Pregnancy
- Some water and heartburn pills, inhalers, statins
- Low magnesium, calcium or potassium
- Diabetes, nerve, kidney, liver or thyroid problems

WHAT THE RESEARCH SAYS

HELPED (1 trial each) — **Stretches and stockings.** Adults over 55 who stretched calves and hamstrings nightly had fewer, milder night cramps; calf stretches alone did not. Knee-high compression stockings by day helped people with good circulation (2026).

EARLY (1 trial) — **Vitamin K2 (MK-7).** One trial (adults 65+, 180 mcg nightly) found fewer night cramps. Its report was reissued with fixes in 2025; it needs a second trial.

NOT PROVEN — **Magnesium.** Unlikely to prevent ordinary night cramps in older adults (Cochrane 2020; a 2026 trial agreed). But low magnesium can itself cause cramps, and some water pills and heartburn pills lower it.

MIXED — **CoQ10 with a statin.** A 2025 review found a little less muscle pain. A 2020 review found no clear benefit.

Hallegraeff 2012 · Coppin 2005 · Kuusipalo 2026 · Tan 2024 (notice 2025) · Cochrane 2020 · Kovacic 2025 · Kennedy 2020

⚠ GET CHECKED

- A sore calf that is **swollen, warm or red. It can be a blood clot. Get care today.**
- On a statin, with **unexplained muscle pain or weakness**, or **dark urine. Call your prescriber now.**
- **Calf pain when you walk** that eases when you stop. **Have your circulation checked.**
- Cramps that come **often**, or with **weakness or numbness. Have a doctor check them.**

Quinine pills are no longer advised for leg cramps.

Lifestyle Recommendations

Bedtime stretch
Gently stretch your calves and the backs of your thighs every night before sleep.

Drink enough
Sip water all day. After long, sweaty workouts, add electrolytes, not just water.

Eat your minerals
Magnesium: greens, beans, nuts and seeds. Potassium: potatoes and bananas.

Build up slowly
Add exercise a little at a time. Tired muscles cramp more easily.

Stop a cramp
Straighten the leg, pull your toes toward you, rub the knot, add warmth.

Check medicines
Water pills, heartburn pills, inhalers and statins can play a part. Ask before stopping.

Nutrient Solutions

What your body needs, and the NutriDyn formula that delivers it

START HERE

Relaxed muscles
magnesium + calcium, theanine, passion flower



Muscle Eze
90 capsules · 30 servings

- Helps muscles relax ◆
- Supports calm and rest ◆

IN EACH SERVING
Per 3 capsules: 200 mg magnesium (citrate), 100 mg calcium (lactate), 50 mg L-theanine, 40 mg passion flower extract

3 capsules at bedtime
may cause drowsiness

Calm and relaxed
magnesium, taurine, B6, L-theanine



Mag Calm Pro
About 60 servings · Raspberry Lemon

- Supports relaxation ◆
- Supports restful sleep ◆

IN EACH SERVING
Per scoop: 200 mg magnesium (malate), 500 mg taurine, 2 g myo-inositol, 100 mg L-theanine, 5 mg vitamin B6

1 scoop a day in water
in place of Muscle Eze

Bone calcium
vitamin K2 as MK-7



Vitamin K2
30 capsules · Progressive Professional brand

- Helps activate bone proteins that bind calcium ◆
- Long-acting MK-7 form

IN EACH SERVING
100 mcg vitamin K2 as menaquinone-7 (MK-7) per capsule

1 capsule daily
with a meal

Muscle cell energy
CoQ10 as ubiquinol



CoQ10 Ubiquinol
30 softgels

- Helps muscle cells make energy ◆
- Ready-to-use ubiquinol form

IN EACH SERVING
100 mg ubiquinol, the ready-to-use form of CoQ10, per softgel

1 softgel daily
with a meal

Mineral balance
magnesium + potassium



Mag-K
90 capsules · Progressive Professional brand

- Supports normal muscle and nerve signals ◆
- Supports healthy mineral levels ◆

IN EACH SERVING
Per 3 capsules: 200 mg magnesium, 99 mg potassium

1 capsule with each meal
in place of Muscle Eze

YOUR DAILY RHYTHM

Evening meal	Vitamin K2 1 capsule
Bedtime	Muscle Eze 3 capsules

The full plan is **4 pills a day**. Start with Muscle Eze; add Vitamin K2 after a week. Swap magnesium products, never stack them.

PHARMACIST CHECK

Keep taking your medicines.

- **Bring your medicine list.** Your pharmacist can check it: some water pills, heartburn pills, asthma inhalers, statins and bone or estrogen medicines are linked to cramps. Never stop one on your own.
- **One magnesium product at a time:** Muscle Eze, Mag Calm Pro or Mag-K. Too much gives loose stools.
- **Kidney disease?** Ask your prescriber before any magnesium product. **On an ACE inhibitor or ARB, or spironolactone, triamterene or amiloride?** Skip Mag-K unless your prescriber says yes.
- **On warfarin (Coumadin)?** K2 and CoQ10 affect it. Start neither without your prescriber.
- **On a statin?** Ask your pharmacist if CoQ10 fits you.
- **Pregnant, nursing, or on sleep or anxiety pills?** Ask your pharmacist first. Muscle Eze can make you drowsy.
- **Thyroid pills:** 4 hours apart from magnesium or calcium. Take **alendronate, doxycycline or ciprofloxacin** 2 hours before magnesium or calcium, or 6 hours after.

Special considerations for you:

Recommended by _____ Phone _____

◆ These statements have not been evaluated by the Food and Drug Administration. These products are not intended to diagnose, treat, cure, or prevent any disease.

NutriDyn